

Part 8: Regression Topics

Heteroskedasticity

To this point, an assumption of our model has been that all of the model errors have the same variance; this condition is called **homoskedasticity**.

The opposite case, the presence of **heteroskedasticity** is a serious concern. The validity of the confidence intervals, hypothesis tests, and so forth rests on this assumption.

The plot of residuals versus fitted values is a useful tool for diagnosing heteroskedasticity.

Exercise: What shapes would one expect to see in the plot of residuals versus fitted values when heteroskedasticity is present?

There are two major strategies for dealing with heteroskedasticity.

First, the response variable can be **transformed**, meaning you would use $g(y)$ instead of y as the response variable. The two most popular choices for $g()$ are the logarithm, and the square root.

There is an accompanying Jupyter notebook that shows a simple example in which the log transformation helps with heteroskedasticity.

The other major strategy for dealing with heteroskedasticity is to use **weighted least squares**. Briefly stated, instead of minimizing the residual sum of squares, one minimizes

$$\text{WRSS}(\beta) = \sum_{i=1}^n w_i (y_i - \hat{y}_i)^2$$

where the weights are chosen to deemphasize those observations for which the variance is larger.

In fact, the optimal choice is $w_i = 1/\sigma_i^2$, where σ_i^2 is the variance for the i^{th} model error.

One would not typically know the σ_i^2 , but it is sometimes the case that you are willing to assume that $\sigma_i^2 = \sigma^2/m_i$. Then, the weight can be chosen as $w_i = m_i$. Or, the σ_i^2 could be estimated from the residuals.

Exercise: Explain why it may be the case that you would be willing to assume that $\sigma_i^2 = \sigma^2/m_i$.

Influential Observations

One source of concern is the possibility that one (or a few) observations are unduly **influential** on the resulting model fit. Of course, we expect that each data point should have some effect on the resulting fit. But no one observation should be overly influential; we'd prefer that each contribute a roughly equal amount to determining the final estimates.

A natural way of quantifying the influence of observation j is to consider the following question:

By how much do the fitted values $\hat{Y}_1, \hat{Y}_2, \dots, \hat{Y}_n$ change if observation j is excluded from the training set, i.e., it is not used in fitting the model?

This question is answered by **Cook's Distance**.

First, let $\hat{Y}_{i(-j)}$ denote the fitted value for observation i when using the model that **excludes** observation j from the fitting. Cook's Distance for observation j is then

$$D_j = \sum_{i=1}^n (\hat{Y}_i - \hat{Y}_{i(-j)})^2 / (p+1)\hat{\sigma}^2$$

1 In Python, we can find Cook's Distance using

2 `holdlm.get_influence().cooks_distance[0]`

3 where `holdlm` is the output from the `statsmodels` function used previously.
4

5 Fortunately, calculating Cook's Distance for an observation does not actually require refitting the model with that observation removed. It is possible to show that (in the case of simple linear regression),

$$D_j = \frac{\hat{\epsilon}_j^2}{(p+1)\hat{\sigma}^2} \left[\frac{h_{jj}}{(1-h_{jj})^2} \right]$$

9 where h_{jj} is the previously-defined i^{th} leverage.

10 **Exercise:** Consider the given expression for D_j . What does it say about
11 what it takes to make an observation influential?

1 Handling Influential Observations

2 Influential observations are almost always an **outlier** in some sense; this is
3 what causes them to have more influence than other observations.

4 First check to see if the influential point is not the result of a mistake. If
5 so, the error should be corrected or, if that is not possible, the observation
6 removed. But, **it is a mistake to arbitrarily remove an observation from a**
7 **training set**. Instead, consider the following:

- 8 1. One should verify that there truly is a linear relationship between the
9 response and predictor. A nonlinear model may be able fit more flexi-
10 bly to the feature(s) that led to this outlier.
- 11 2. It may be the case that this observation is sufficiently different from the
12 others that it would make sense to redefine the population of interest
13 in such a way that this outlying (and other) cases are handled by a
14 separate model. In other words, it may not be realistic to expect a
15 single model to cover such a wide range of possibilities.
- 16 3. A transformation of the predictor and/or response can "pull in" an
17 outlier so that it's not so extreme.
- 18 4. An alternative approach to estimating the β_i may be preferable; for
19 example, weighted least squares or robust regression.

Making Predictions

One of the fundamental motivations for using regression models is to make predictions for **future** observations of the response. In other words, consider a situation where you know the value of the predictor variable x , and seek to predict the response.

In the context of our simple linear regression model, if our value for the predictor is x^* , then we would start by finding

$$\hat{y}^* = \hat{\beta}_0 + \hat{\beta}_1 x^*$$

We have plugged in this new value for x into the line that has been fit. This is our best estimator for y^* , the true response for this object.

One should not be satisfied with only this, however: We should quantify the amount of error in this prediction.

In the case of simple linear regression, the **variance in the error in the prediction** is

$$V(\hat{y}^* - y^*) = \sigma^2 + \sigma^2 \left[\frac{1}{n} + \frac{(x^* - \bar{x})^2}{(n-1)s_x^2} \right]$$

Exercise: Explain how the above decomposes the error in the prediction into two key sources.

Hence the **standard error in the prediction** is

$$SE(\hat{y}^*) = \sqrt{\sigma^2 + \sigma^2 \left[\frac{1}{n} + \frac{(x^* - \bar{x})^2}{(n-1)s_x^2} \right]}$$

Of course, in practice σ^2 will be replaced with $\hat{\sigma}^2$.

- 1 If the errors are assumed to be normally distributed, then a $100(1 - \alpha)\%$
2 **prediction interval for the future response** can be written as

$$\hat{y}^* \pm t_{\alpha/2, n-2} \text{SE}(\hat{y}^*)$$

- 4 Figure 1 shows a simple example of how the prediction interval will ap-
5 pear, as x^* ranges over the extent of the predictor space.

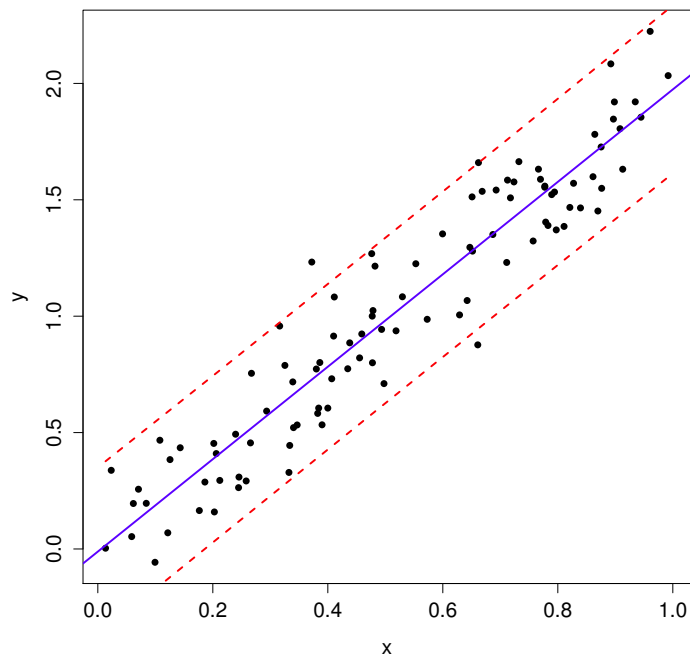


Figure 1: An example showing a regression line (solid line), along with the prediction bands (dashed lines).

- 1 **Exercise:** Discuss the role of the normality assumption in the validity of
2 this interval. If the sample size is large, do you believe that the normality
3 assumption can be relaxed?

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Discrete Variables as Predictors

One often encounters variables which are discrete quantities, but still are on the ratio scale.

A primary example is that of a **count variable**, i.e., a variable which counts the number of times something occurs. For example, it is standard for credit-granting entities to build models where the response variable is

- the likelihood someone will payoff a loan

and the predictors utilized include

- number of major derogatory reports on credit record
- number of dependents
- number of major credit cards held
- number of active credit accounts

Such predictors **can** be included as predictors just as continuous quantities are; the interpretation remains that same: The expected response increases at a constant rate as the variable increases.

Categorical Predictors

We just discussed using a discrete variable on the ratio scale as a predictor. There are some discrete variables for which this is clearly not a good idea, however. These are cases where the numbers are merely labels for the levels of a **categorical variable**.

Consider a variable which indicates highest degree obtained:

Code	Value
0	Did not finish high school
1	Finished high school
2	Some college
3	Graduated college

Exercise: This variable is on the ordinal scale. What would happen if we mistakenly treated this as a variable on the ratio scale when including it in the regression model?

1 **Here is the tricky part:** A categorical variable with k levels adds $k - 1$
2 terms into the regression function.

3 To see why this is the case, imagine that the model under consideration
4 includes a predictor giving the number of credit cards, along with this ed-
5 ucation level predictor. The $k - 1$ new terms correspond to “shifts” in the
6 regression function relative to the “baseline” category. In particular, let X_1
7 equal the number of credit cards. Then,

when education level is . .	the regression function is . . .
0	$\beta_0 + \beta_1 X_1$
1	$\beta_0 + \beta_1 X_1 + \beta_2$
2	$\beta_0 + \beta_1 X_1 + \beta_3$
3	$\beta_0 + \beta_1 X_1 + \beta_4$

9 We can write this regression function more compactly as

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$$\beta_0 + \beta_1 X_1 + \beta_2 \mathbf{1}_{\{\text{edu} = 1\}} + \beta_3 \mathbf{1}_{\{\text{edu} = 2\}} + \beta_4 \mathbf{1}_{\{\text{edu} = 3\}}$$

11 where $\mathbf{1}_A$ is the **indicator variable** for the event A .

12 **Exercise:** Why not include **every** discrete predictor as being categorical?

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1 An appropriately-defined categorical variable will be handled correctly by
2 software such as Python and R.

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3 Coefficients:
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5      Estimate Std. Error t value Pr(>|t|)
6 (Intercept)    35.2647    0.7990   44.137 < 2e-16 ***
7 NumCards      -2.1467    0.1915  -11.209 < 2e-16 ***
8 (Education)1    5.4004    0.8424    6.411 5.58e-09 ***
9 (Education)2    7.8708    0.9308    8.456 3.26e-13 ***
10 (Education)3    9.5610    0.8292   11.530 < 2e-16 ***
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10 **Exercise:** Consider the sample output above. What is the expected re-
11 sponse for a case when education level is “Did not finish high school?”
12 What about when education level is “Some college?”

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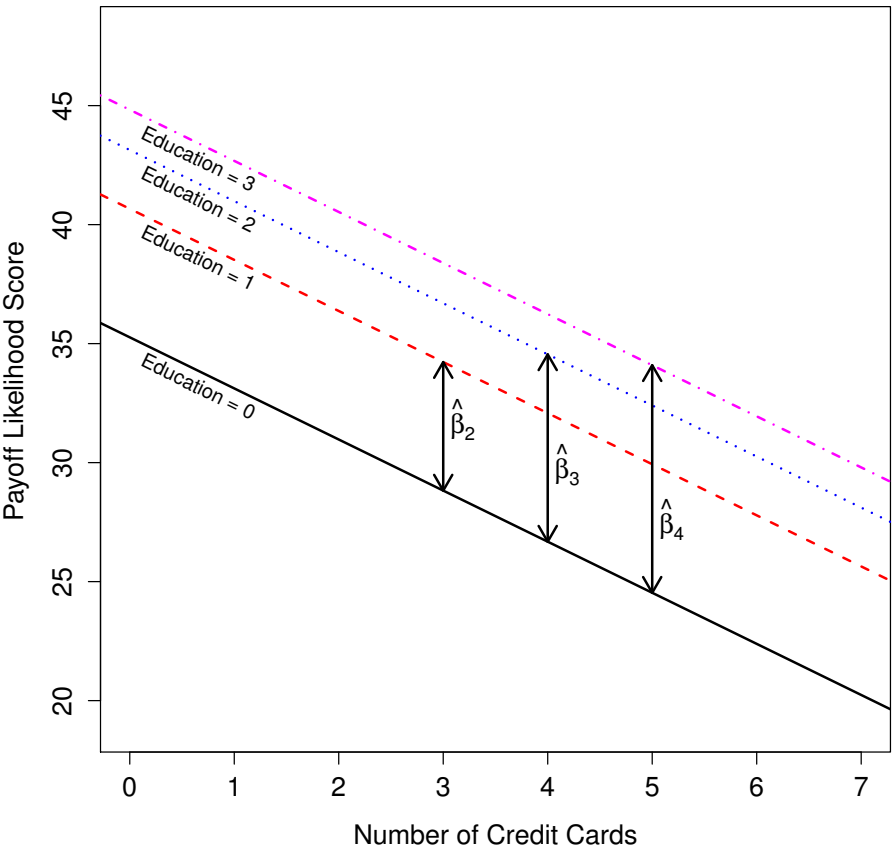


Figure 2: The fit model in this case. The four lines correspond to the four levels of the Education predictor. The slope of each line is $\hat{\beta}_1 = -2.15$.

Categorical Variables in the Design Matrix

Remember that a categorical variable with k levels adds $k - 1$ β parameters to the model. Hence, it adds $k - 1$ **columns** to the design matrix $\mathbf{X}$. In order to create the “shifting” effect, these columns will be filled with zeros and ones.

Consider the following example. The response is the assessment of the likelihood of default. There is a predictor which counts the number of open credit cards. There is another predictor “education level,” defined as above. The first few rows of the data set appear as follows:

Observation	Payoff Likelihood	Number of Cards	Education Level
1	40	4	2
2	41	2	3
3	44	1	2
4	28	5	0
5	36	4	1
6	32	2	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$

Assume that the analysis proceeds with the predictor “Number of Cards” untransformed.

1 The first six rows of the design matrix would be

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$$\mathbf{X} = \begin{bmatrix} 1 & 4 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 5 & 0 & 0 & 0 \\ 1 & 4 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

3 **Exercise:** Explain the form of the design matrix above. Consider the matrix
4 product $\mathbf{X}\beta$.

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1 **More than One Categorical Variable**

2 **Exercise:** Suppose that a model contains only two predictors, and they are
3 both categorical variables. The first (X_1) has three levels, and the second
4 (X_2) has four levels. How many total β parameters are in the model, and
5 how is each interpreted?

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Interactions

Interaction terms are added to the model by taking the product of other predictors.

In the absence of any interactions, we say that the model is **additive** in the predictors.

Exercise: Think carefully about what is being assumed when an additive model is used. In particular, what is the effect on the regression function if a covariate is varied?

Consider a model with two predictor variables X_1 and X_2 , but with the interaction term X_1X_2 included. So, the regression function is

$$\beta_0 + \beta_1X_1 + \beta_2X_2 + \beta_3X_1X_2$$

Then, if X_1 is held fixed, the effect of varying X_2 is characterized by the slope of $\beta_2 + \beta_3X_1$. In other words, if X_2 is increased by c , then the regression function increases by $(\beta_2 + \beta_3X_1)c$.

The key enhancement is that this slope now depends on X_1 .

Of course, one may wonder: Why does it make sense to let the slope depend on X_1 **in this particular manner**?

The best way to answer this is to consider the interaction X_1X_2 to be a **first order approximation** to this dependence.

It is much like how we often use straight lines to approximate relationships which are not truly straight: These are useful approximations in the right circumstances.

1 **Exercise:** What will happen if you include in the model an interaction be-
2 tween a continuous predictor and a categorical variable?

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